Bubble Sort Algorithm

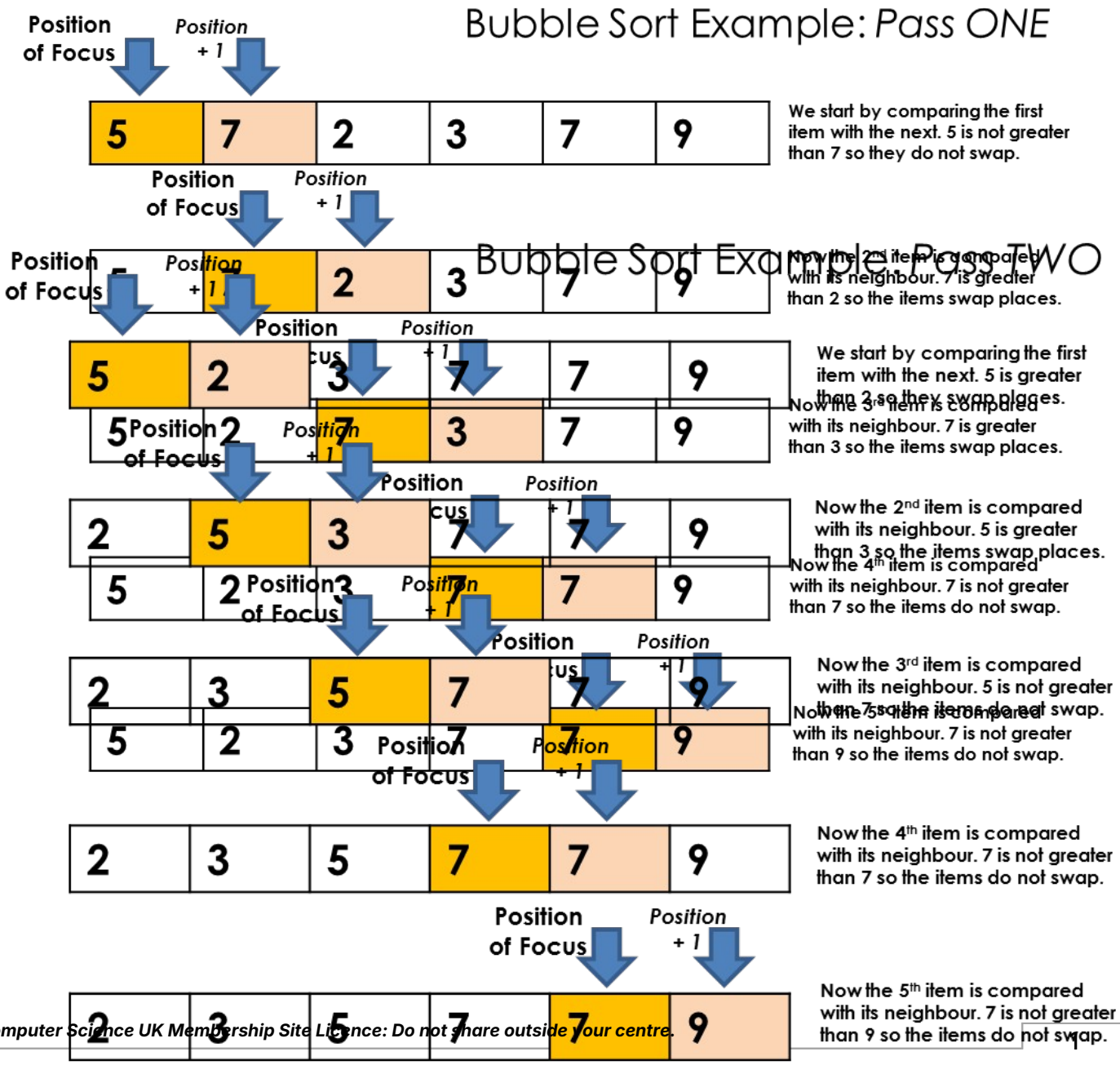
Sort Algorithms

Sorting algorithms are used to sort data in particular ways (for example sorting a set of numbers from highest to lowest). There are some standard sorting algorithms that have been developed and each have their own merits. Over the next few pages we will take a look at the 'Bubble Sort' algorithm.

The Bubble Sort Algorithm

A bubble sort is quite simple in the way it works. It repeatedly steps through a list of data, comparing each pair of data it comes across and swaps them if their order is wrong.

An Example:



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It is important to recognise that although the above dataset is now in order, there will be a third and final pass carried out on the dataset. This is because algorithms are not intelligent, they cannot see that the dataset has been sorted until a pass is carried out in which no swaps are made.

Bubble Sort Algorithm Pseudocode Example:

```
WHILE unsorted == True:
        unsorted = False
        FOR position 1 to lengthOfdataset:
                IF dataset[position] > dataset[position+1]
                        unsorted = True
                        temp = dataset[position]
                        dataset[position] = dataset[position+1]
                        dataset[position+1] = temp
```

Questions *(The question zone you choose must either match your target grade or be higher!)*

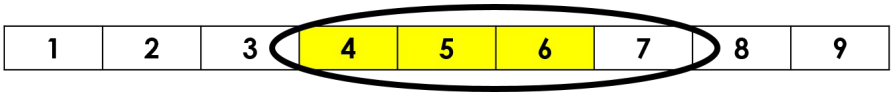
1	2	3	4	5	6	7	8	9

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Question Zone 1-3

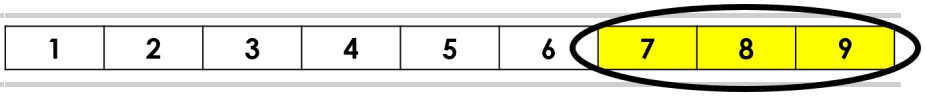
1. What is the purpose of the 'Bubble Sort Algorithm'?. [1]
2. What does the term 'pass' mean, when discussing 'Bubble Sort Algorithms'? [3]
3. Describe how the 'Bubble Sort' works. [3]

Question Zone 4-6



1. Describe how the 'Bubble Sort' works. [3]
2. For a given dataset, a 'Bubble Sort Algorithm' takes 4 passes to sort a set of data. Will the algorithm stop at this point? Explain your answer. [3]
3. Show the steps that will take place to sort the following dataset, during a 'Bubble Sort'. [5]

Questions Zone 7-9



1. Show the steps that will take place to sort the following dataset, during a 'Bubble Sort'. [5]

[10, 5, 20, 15, 0]

2. How many comparisons and swaps are required to sort the following set of data using a 'Bubble Sort'? You must show your working out. [4]

[2, 5, 4, 8, 7, 6]

3. What is the maximum number of comparisons and swaps that a 'Bubble Sort Algorithm' could make if there are 5 items of data in a dataset? You must show your working out. [5]

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Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Handwriting neat & legible
- Punctuation / Capital letters

Keywords / Key Terms:

Bubble Sort: *A method of sorting a set of data by repeatedly stepping through the dataset and swapping any pairs that are in the wrong order.*

A Pass: *A pass is the name given to describe the algorithm 'making its way through' the dataset during each iteration of a bubble sort algorithm.*

Algorithm*: A set of steps/instructions needed to complete a software task.*

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Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples**: Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**!

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Reflections **Score:** / **Percentage:** % **Grade:** **Progress** On / Above / Below

What Went

- ☐ I demonstrated a good level of understanding.
- ☐ I responded to the command words effectively.
- ☐ My answers were detailed / were written in depth.
- ☐ My work was well presented / legible.
- ☐ My answers effectively incorporated technical terminology.
- ☐ My responses were well structured / organised.
- ☐ My revision strategy was effective as I showed depth of understanding in my answers.
- ☐ My answers contained enough points / examples / explanations to achieve the marks available.

Even Better

- ☐ My answers need to be more accurate.
- ☐ I must respond correctly to the command words.
- ☐ My answers need more detail / greater depth.
- ☐ I must take greater care over my work / write neatly.
- ☐ I must incorporate key terminology into my answers.
- ☐ I must better organise my answers to improve its clarity.
- ☐ I need to improve my revision strategy, as I did not demonstrate a depth of understanding in my answers.
- ☐ My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts:

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